B.M.S. COLLEGE OF ENGINEERING

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DEPARTMENT OF MACHINE LEARNING

(UG Program: B.E. in Artificial Intelligence and Machine Learning)

Course : Natural Language Processing for LLM
Course Code: 24AM7PCLLM

Comparative Evaluation of Prompt Engineering Techniques for Structured Clinical Information Extraction Using Local Large Language Models

Alternative Assessment Tool (AAT)

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Agenda

- Introduction
- Selected NLP Topic Overview
- Real-Time Application / API Identified
- NLP Model / Architecture Study
- Dataset and Preprocessing
- Implementation Methodology
- Results and Evaluation
- Discussion
- Conclusion
- References

Project at a glance

5
prompt strategies compared

40
inference runs (8 cases × 5 strategies)

1.5B
parameter model, run fully locally

100%
on-device: no patient text leaves the machine

Introduction and Objectives

1980s–1990s
Rule-based
hand-crafted rules; brittle on colloquial text

→

1990s–2010s
Statistical ML
HMMs, CRFs, TF-IDF features

→

2013–2017
Deep learning
Word2Vec, GloVe, LSTM / BiLSTM-CRF

→

2017–present
Transformers, LLMs
attention [3], BERT [6], GPT-3 [1], Qwen

Why clinical extraction?

- **~80%** of clinical information is free-text narrative (often-cited estimate)
- EHRs and HL7 FHIR [7] need structured records; decision support and cohort search cannot use free text
- HIPAA / GDPR limit sending PHI to the cloud; a local 1B–3B model avoids that but is unreliable at structured output

Objectives and scope

- Benchmark **five prompt strategies**
- Run **fully locally**: Qwen2.5:1.5B on Ollama
- Evaluate on **many criteria**: validity, schema coverage, field F1, negation, latency
- Streamlit app for live comparison
- Scope: first **8 of 20** synthetic cases (40 runs), one model, research prototype

Research question: Which prompt technique best closes the structured-output gap of a small local LLM?

The NLP Task: Structured Extraction

Task: turn a free-text clinical note into a fixed JSON schema (EHR / HL7 FHIR fields). Classical NLP needed four stages (NER → negation → relation extraction → slot filling); an LLM does it in **one prompted step**, producing the JSON token by token.

Real-world use cases

- **EHR ingestion** from dictated notes
- **Decision support:** flag contraindications and interactions
- **Medical coding** and billing (ICD-10, CPT)
- **Trial matching** from structured registries
- **Pharmacovigilance:** spot adverse drug events

Key challenges and how we measure them

Challenge	Metric
Syntactic fragility	Strict RFC 8259 validity
Clinical negation	Negation violation rate
Schema hallucination / key drift	Schema key coverage
Multi-attribute entity linking	Medications F1
Latency versus reasoning	Latency and token counts

Why it is hard: 1B–3B models often emit unquoted units (198 mg/dL), trailing commas or chatty preambles that break json.loads, and "denies chest pain" must never become an active symptom [8].

Application: Ollama API and Streamlit

Streamlit app (app.py) · **CLI runner** (run_experiment.py)

↓

Prompt strategy layer
Zero-Shot · Few-Shot · CoT · Schema · Role-Expert

↓

LocalLLMClient → Ollama server
HTTP POST to :11434 · qwen2.5:1.5b (GGUF Q4_K_M)

↓

Self-healing JSON parser → StructuredIEEvaluator

↓

Results: JSON logs · CSV summary · five plots

Inference backend: Ollama [15]

Local REST API (localhost:11434) serving the quantized qwen2.5:1.5b. Nothing is streamed to external servers.

User-facing app: Streamlit [16]

Clinical Extraction Studio and Benchmark Explorer, with four modules:

- Live Extraction Playground
- Comparative Benchmark Dashboard
- Prompt Engineering Theory & Prompts
- Project Report & Architecture

Input: note + strategy + temperature (0.1). **Output:** JSON (8-field schema) + latency and token counts.

Qwen2.5-1.5B-Instruct: Architecture

Tokenizer + embedding
byte-level BPE (~151k tokens) → 1536-d vectors

Decoder block, repeated 28 times

RMSNorm → **grouped-query attention**
(causal, RoPE) + residual add

RMSNorm → **SwiGLU feed-forward**
+ residual add

Final RMSNorm → LM head → softmax
chosen token is appended, K/V cached, loop repeats

Stops at the end token or the 1024-token limit

1.54 B parameters	**28** decoder layers	**12 Q / 2 KV** attention heads (GQA)
32k native context (4096 used)	**~1 GB** quantized (Q4_K_M)	**18 T** pre-training tokens

Why it suits extraction

- Strong instruction following and JSON output [4]
- About 1 GB quantized: runs on CPU-only laptops
- Fully on-premise: supports HIPAA and GDPR goals
- A demanding test bed: prompt effects are measurable

$$\text{Attention}(Q, K, V) = \text{softmax}\left(QK^T / \sqrt{d_k} + M \right) V$$

Few-shot adds K examples; CoT: $P(R \mid X, I) \cdot P(Y \mid X, R, I)$

Dataset and Preprocessing

20
synthetic encounter cases

2
held-out exemplars, used only in prompts

8
medical specialties

0
real patient records or PHI

Evaluated: first 8 cases → 8 × 5 = 40 runs. Frozen weights, so no train/test split.

Gold annotation, diabetic neuropathy case (abridged)

```
{
  "patient_age": 62,  "patient_gender": "Male",
  "primary_diagnosis": "Type 2 Diabetes
     Mellitus with Peripheral Neuropathy",
  "severity_level": "Moderate",
  "symptoms": ["bilateral burning foot pain",
     "paresthesias", "polyuria", ...],
  "medications": [{"name": "Metformin",
     "dosage": "1000 mg",
     "frequency": "twice daily"}, ...],
  "vitals_and_labs": {"HbA1c": "9.4%", ...},
  "follow_up_days": 30
}
```

Cleaning the model output: self-healing JSON parser

1. **Fence stripping** — take the inside of json code blocks
2. **Boundary scan** — keep text between the outer { and }
3. **Unit repair** — 198 mg/dL → "198 mg/dL"
4. **Trailing commas** — delete commas before a closing brace

Strict validity is judged on the raw output, so repair can rescue scoring without hiding model failure.

Five Prompt Engineering Strategies

Strategy	Core idea	Prompt tokens
Zero-Shot	Direct instruction to extract the information; no schema, no examples	247
Few-Shot ICL [1]	Two held-out input–output demonstrations show the keys, nesting and quoting	971
Chain-of-Thought [2]	Reasons step by step about the note, including negations, then emits the JSON	948
Schema-Constrained	Formal schema with types, required keys and allowed values, plus strict validity rules	534
Role-Expert	"Principal Clinical Informatician" persona with guardrails: no hallucination, negation filtering	607

Same pipeline for all five: only the prompt text differs



Settings: T = 0.1 · top-p 0.9 · 1024 new tokens · 4096 context · K = 2 exemplars · 40 runs · weights frozen

Evaluation Metrics

Structural

Strict JSON validity (RFC 8259 [14], checked before any repair) and **schema key coverage** (share of the 7 scored keys).

Accuracy

Field-level F1: demographics, diagnosis, severity, symptoms (P / R / F1), medications and follow-up days.

Safety

Negation violation rate: share of extracted symptoms that are actually negated in the note, e.g. “denies fever” [8].

Cost

Latency per sample, prompt and completion **tokens**, and **tokens per second**.

Overall F1 (composite; weights are a design choice and were not tuned)

$$F1_{overall} = \max(0,\ 0.15\,F1_{demo} + 0.25\,F1_{diag} + 0.10\,F1_{sev} + 0.25\,F1_{symp} + 0.20\,F1_{med} + 0.05\,F1_{follow} - 0.15\,R_{neg})$$

Free-text fields: Precision = $|pred \cap gold| / |pred|$, Recall = $|pred \cap gold| / |gold|$, $F1 = 2 \cdot P \cdot R / (P + R)$

Benchmark Results (n = 8 Cases)

Strategy	Strict valid	Schema cov.	Diag F1	Symp F1	Meds F1	Overall F1	Neg. viol.	Latency
Few-Shot ICL	**100.0%**	98.2%	**83.9%**	82.9%	**100.0%**	**91.1%**	**0.0%**	13.1 s
Schema-Constrained	**100.0%**	**100.0%**	80.8%	78.9%	97.5%	89.4%	**0.0%**	**12.1 s**
Chain-of-Thought	**100.0%**	96.4%	73.5%	**86.3%**	**100.0%**	88.1%	**0.0%**	20.8 s
Role-Expert	87.5%	87.5%	55.4%	65.5%	87.5%	70.6%	1.4%	13.3 s
Zero-Shot †	75.0%	1.8%	0.0%	0.0%	0.0%	0.0%	12.5%	15.3 s

Few-Shot ICL
Highest overall F1 (91.1%) and best diagnosis F1

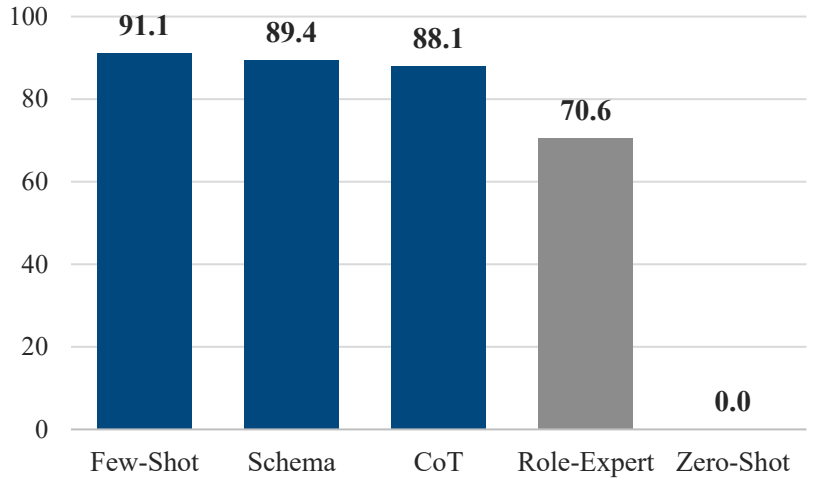
Schema-Constrained
Only strategy at 100% on both structure measures, and the fastest (12.1 s)

Chain-of-Thought
Best symptoms F1 (86.3%), but about 72% slower (20.8 s)

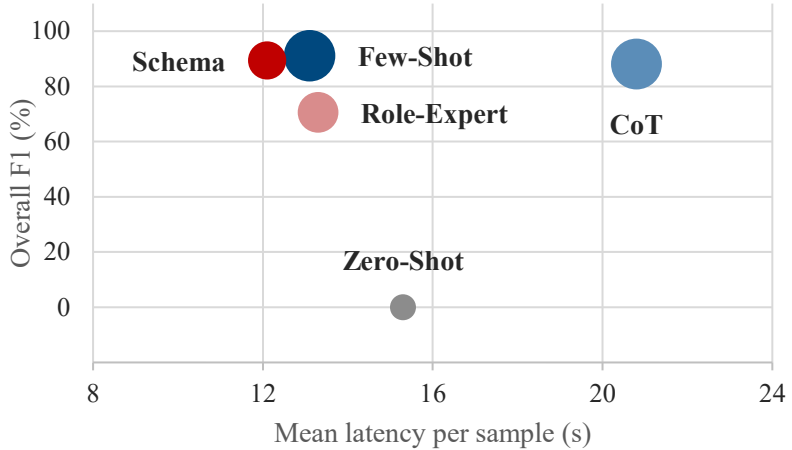
Red on tinted cells = best in column; means over 8 cases. † Zero-Shot is scored against a schema it is never told, so its zeros mostly measure schema discovery, not clinical understanding.

Accuracy versus Cost

Overall F1 (%), mean of 8 cases



Latency vs overall F1 (bubble = prompt tokens)



Few-Shot, Schema and CoT are within 3 points. With 8 cases, one case moves a mean by up to 12.5 points.

Zero-Shot: 75% strictly valid JSON, yet only 1.8% of the expected keys appear.

Latency follows generated text, not prompt size: Few-Shot has 82% more prompt tokens than Schema but is ~1 s slower; CoT is ~8 s slower.

Interpreting the Results

1 Structure is the first bottleneck

Once the format is given, the 1.5B model fills most fields correctly (three strategies above 88% F1). Without it, the target structure never appears.

2 Two ways to supply structure work

Exemplars (Few-Shot) and an explicit schema give almost the same accuracy; the schema needs about half the prompt tokens.

3 Reasoning helps one field, at a price

CoT has the best symptoms F1, but Few-Shot and Schema also reach 0.0% negation violations. CoT costs 72% more latency.

4 Persona is not a substitute for structure

Role-Expert added a longer prompt and guardrails, yet trailed Schema-Constrained by about 19 points of overall F1.

Strengths

- Fast and cheap: 12–13 s per note, no API fees
- 100% strict validity for three strategies
- Complete data privacy: nothing leaves the machine

Limitations

- Only 8 cases: one case shifts a mean by 12.5 points
- One run per pair, synthetic notes, one model, 4096-token context
- Free-text diagnosis wording is the hardest field

Conclusion and Future Work

Best overall trade-off: Schema-Constrained prompting. 100% strict validity and coverage, 89.4% F1, 12.1 s per note, 534-token prompt. Few-Shot (91.1% F1) is a strong choice when prompt length is not a constraint.

Key learnings

1. **Zero-Shot is unsuitable:** valid-looking JSON, wrong structure (1.8% coverage, 0.0% F1)
2. **Few-Shot has the highest F1** (91.1%), with 82% more prompt tokens for a gain within noise
3. **CoT improves symptoms** (86.3% F1) at about 72% more latency
4. **Prompt structure matters more than persona**
5. **Report limitations with results:** one case moves a mean by 12.5 points

Future work

- **Constrained decoding:** GBNF, Outlines or Ollama JSON-schema output
- **Stronger evaluation:** all 20 cases, repeats, confidence intervals
- **Hybrid prompts:** schema + 1–2 exemplars + short reasoning
- **Negation post-filter:** NegEx-style rules [8]
- **LoRA fine-tuning** on de-identified corpora (MIMIC-IV)
- **Larger models:** 3B, 7B and 14B on GPU

Before any clinical use: validate on real de-identified notes, schema-check every output, add human review.

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Thank you !